

Testing bandwidth and latency

iperf: iperf is a network performance measurement tool that tests bandwidth and identifies performance issues between two hosts.

Usage benefits	Example command
Measures TCP bandwidth	<code>iperf -c server_ip</code>
Measures UDP bandwidth	<code>iperf -u -c server_ip</code>
Runs as a server for testing	<code>iperf -s</code>

mtr: mtr (My Traceroute) combines the functionalities of ping and traceroute to provide real-time network diagnostics.

Usage benefits	Example command
Traces route and measures latency	<code>mtr google.com</code>
Displays packet loss per hop	<code>mtr -r google.com</code>
Runs in report mode	<code>mtr -w google.com</code>

Network diagnostics

nc (netcat): nc (Netcat) is a versatile networking tool used for debugging and testing. It can establish connections, listen to ports, and transfer data.

Usage benefits	Example command
Tests port connectivity	<code>nc -zv host_ip port</code>
Listens on a specific port	<code>nc -l port</code>
Sends data to a remote host	<code>`echo "message"   nc host_ip port`</code>

arp-scan: arp-scan identifies active devices on a local network by sending ARP requests to all IP addresses.

Usage benefits	Example command
Scans the local network	<code>arp-scan -l</code>
Targets a specific IP range	<code>arp-scan 192.168.1.0/24</code>
Detects unauthorised devices	<code>`arp-scan -l grep unauthor-ized`</code>

By integrating these tools into daily operations, administrators can maintain high-performing and reliable network environments.

Monitoring and visualisation tools

Monitoring and visualisation tools are indispensable for network administrators, offering insights into network performance and trends over time. These tools enable administrators to track traffic, identify anomalies, and create dashboards for a holistic view of the network's health and efficiency.

Key questions administrators face daily in this respect are:

- Is network traffic usage within expected limits?
- Are there any trends or anomalies in network performance that require attention?

Traffic monitoring

vnstat: vnstat is a lightweight, command-line-based network traffic monitor that logs data and provides historical bandwidth usage statistics.

Usage benefits	Example command
Tracks historical bandwidth usage	<code>vnstat</code>
Displays detailed daily and monthly stats	<code>vnstat -d</code>
Shows real-time traffic updates	<code>vnstat -l</code>

bmon: bmon (Bandwidth Monitor) is a visual, real-time traffic monitoring tool that displays bandwidth usage per interface.

Usage benefits	Example command
Visualises bandwidth usage in real-time	<code>bmon</code>
Monitors traffic on specific interfaces	<code>bmon -p eth0</code>
Logs data for later analysis	<code>bmon -o logfile</code>

Visualisation platforms

Grafana: Grafana is an open source visualisation platform used to create interactive and customisable dashboards for monitoring network and system performance.

Usage benefits	How to use it
Creates interactive dashboards	Access via web interface (default: <code>http://localhost:3000</code>)
Integrates with various data sources	Configure data sources like Prometheus or InfluxDB
Tracks historical trends and metrics	Use the UI to query and visualise data